Problem Statement

Visually impaired individuals face persistent challenges in accessing effective, affordable, and interactive educational tools for Braille literacy. While traditional Braille systems provide tactile learning, they lack the technological integration necessary for auditory feedback, portability, and engagement. The *Braillear* project addresses these shortcomings through an embedded, touch-based, and audio-assisted learning platform.

1. **Limited Accessibility of Braille Learning Tools**
    - Most Braille learning systems available today are expensive, region-specific, and inaccessible to learners in low-resource settings.
    - Dependence on specialized institutions and trained personnel restricts large-scale adoption among visually impaired students.

2. **Lack of Interactive and Multisensory Feedback**
    - Traditional Braille books provide only tactile input with no corresponding auditory or digital reinforcement.
    - The absence of real-time audio feedback limits cognitive association between characters and phonetic sounds, slowing down learning efficiency.

3. **Dependence on Bulky and Non-Portable Devices**
    - Conventional Braille teaching devices are large, power-dependent, and unsuitable for mobile or personal learning environments.
    - Their non-portability hinders flexible usage across classrooms, homes, and rehabilitation centers.

4. **High Cost and Maintenance Complexity**
    - Existing digital Braille systems rely on mechanical actuators or proprietary modules, increasing the overall production cost.
    - Regular calibration and hardware upkeep raise maintenance challenges, making them unsustainable for continuous educational use.

5. **Lack of Real-Time Audio–Tactile Integration**
    - Current assistive systems do not provide simultaneous tactile detection and auditory feedback for instant recognition.

- Delayed response times and inconsistent data transmission disrupt the learning flow and reduce engagement.

6. **Limited Technological Scalability**
    - Most available solutions lack modularity and cannot integrate newer technologies such as wireless modules or app interfaces.
    - This restricts feature expansion, real-time updates, and compatibility with modern embedded or IoT systems.

7. **Absence of Inclusive and Human-Centered Design**
    - Many systems overlook ergonomic aspects like sensor sensitivity, layout spacing, and comfort for visually impaired users.
    - Lack of adherence to accessibility design principles (e.g., ISO 9241-210) affects usability and learning experience.

8. **Software–Hardware Interfacing Challenges**
    - Inefficient communication between capacitive sensors and microcontrollers leads to false triggers or input loss.
    - Serial communication via UART often experiences data lag, affecting audio synchronization and reliability.

9. **Lack of Affordable, Open-Source Educational Platforms**
    - The scarcity of open-source, low-cost assistive learning systems limits innovation in accessible education technology.
    - There is a need for a microcontroller-based, scalable model—such as one using Raspberry Pi Pico and TTP223 sensors—that democratizes Braille learning for all.